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April 23, 1980
SUPERSEDING
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January 27, 1976

FEDERAL SPECIFICATION

SOAP, TOILET (LIQUID AND PASTE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers liquid and paste toilet soaps for use in liquid soap dispensers.

1.2 Classification. Toilet soap shall be of the following types, as specified (see 6.2):

Type I - Liquid.
Type II - Paste.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; Los Angeles; San Francisco; and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications, standards, commercial item descriptions, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 460 - Sampling and Chemical Analysis of Soaps and Soap Products.
D 1544 - Color of Transparent Liquids (Gardner Color Scale).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

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Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Finished product. Type I liquid toilet soap shall be a clear, uniform solution. Type II paste toilet soap shall be a firm, uniform, translucent gel. The toilet soaps shall be mildly perfumed.

3.2 Physical and chemical requirements. Toilet soaps shall conform to the requirements of table I, when tested as specified in 4.3.1.

TABLE I. Physical and chemical requirements

Property	Type I		Type II	
	min.	max.	min.	max.
Anhydrous soap (as potash soap, percent)	15.0	--	52.0	--
Free alkali (as KOH, Percent)	--	0.05	--	0.2
Free acid (as oleic acid, percent)	--	1.0	--	1.5
Matter insoluble in water (percent)	--	0.1	--	0.2
Unsaponified plus unsaponifiable matter (percent)	--	0.6	--	2.4
Acid number (mg KOH/g fatty acids)	215	--	215	--

3.3 Sediment and suspended material. The toilet soaps shall be manufactured or processed so as not to contain any sediment or suspended material.

3.4 Accelerated aging. The soaps shall not become rancid when tested with the accelerated aging test specified in 4.3.2.

3.5 Stability. After freezing and thawing, undiluted type I soap, and a one part of type II soap to three parts water solution, shall reconstitute to a clear solution, when tested as specified in 4.3.4.

3.6 Sudsing. The soaps shall produce a net volume of not less than 100 ml of foam when tested as specified in 4.3.5.

3.7 Sequestering agent (type II). Type II paste toilet soap shall contain sufficient organic sequestering agent so that a clear solution is produced when the product is tested as specified in 4.3.6.

3.8 Color. When tested as specified in 4.3.3, the soaps shall be not darker than Gardner Color Standard No. 10.

3.9 Directions for use (type II). Type II paste toilet soap shall be marked with the manufacturer's normal commercial directions for use and dilution, or with the following:

"Dilute 1:3 with water before placing in dispensers. Measure one (1) quart of soap and three (3) quarts of hot water (approximately 140 degrees F) into a clean container and stir the mixture by hand or with a motor driven stirrer until the paste dissolves completely. (Fifteen to thirty minutes of stirring are usually required.)"

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein.

4.2 Quality conformance inspection.

4.2.1 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one shipping container fully prepared for delivery. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level S-2, acceptable quality level (AQL) of 4.0 expressed in terms of percent defective.

4.2.2 Examination of the toilet soap. The toilet soap shall be examined for compliance with 3.1 and 3.9, and net contents less than specified in the contract. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level S-1, AQL of 2.5 percent defective. The sample unit shall be one unit container of soap.

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4.2.3 Examination of type I liquid toilet soap for sediment and suspended material. The liquid toilet soap shall be examined for sediment and suspended material. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level S-4, AQL of 1.0 percent defective. The sample unit shall be one bottle or drum of liquid toilet soap. With a minimum of agitation, pour 1-2 cups of liquid soap into a clear glass container. Thoroughly agitate the bottle or drum and pour another 1-2 cups of liquid soap into another clear glass container. Examine the two containers. There shall be no sediment or suspended material in the liquid soaps. The agitated liquid soap shall not be cloudier than the unagitated liquid soap (which would indicate finely divided material stirred up from the bottom of the container).

4.2.4 Testing of the toilet soap. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level S-1, AQL of 4.0 percent defective. The sample unit shall be 1 gallon of liquid toilet soap, or 8 pounds of paste toilet soap.

4.3 Test methods.

4.3.1 Analyze the toilet soap for the following properties by the referenced section in ASTM D 460:

4.3.1.1 Free acid - section 18.

4.3.1.2 Matter insoluble in water - section 19.

4.3.1.3 Anhydrous soap - sections 21-22.

4.3.1.4 Free alkali - sections 27-29.

4.3.1.5 Unsaponified plus unsaponifiable matter - sections 30-32.

4.3.1.6 Acid number - sections 45-46.

4.3.2 Accelerated aging. Place 60 ml of the soap in a closable container. Close the container and heat in an oven at a temperature of 65°C (149°F) for 96 hours. Remove from oven and allow to cool to room temperature. There shall be no rancid odor, or other evidence of rancidity.

4.3.3 Color. Test the type I toilet soap and a 1:3 dilution of the type II toilet soap in distilled water by ASTM D 1544.

4.3.4 Stability. Freeze for 8 hours at -5°C (23°F) a 400 ml sample of type I toilet soap or 400 ml of a 1:3 dilution of type II toilet soap in distilled water. Allow to thaw at room temperature overnight. The type I toilet soap and diluted type II toilet soap shall be clear with no clouding or jelling. The diluted type II toilet soap shall not contain sediment or suspended material.

4.3.5 Sudsing. Place 50 ml of undiluted type I soap or diluted type II soap (prepared in 4.3.4) in a 250 ml graduated cylinder. Stopper the cylinder and agitate by inversion (rotating the cylinder about its midpoint without translational motion) for 1 minute at such a rate that 30 inversions are completed. Place the cylinder in an upright position on a table, remove the stopper, wait 5 seconds and read the net volume of foam (total volume minus volume of liquid).

4.3.6 Sequestering agent. Dilute one part of type II soap with three parts water having a hardness equivalent to 500 parts per million calcium carbonate. Mix thoroughly. The resulting solution shall be clear. To prepare 500 ppm hard water, dissolve 0.5545 grams of reagent grade anhydrous calcium chloride in distilled water and dilute to a final volume of 1 liter.

5. PREPARATION FOR DELIVERY

5.1 Packaging (commercial). The toilet soap shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during multiple shipments, handling, and storage.

5.2 Packing (commerical). The packaged toilet soap shall be packed in containers to insure delivery at destination, to provide for redistribution by the initial receiving activity, and shall be acceptable by common carrier under the National Motor Freight Classification and Uniform Freight Classification.

5.3 Marking. Marking shall be as specified in the contract or order (see 6.2).

6. NOTES

6.1 Intended use. The toilet soaps covered by this specification are intended to be dispensed from washroom liquid soap dispensers. Type I liquid toilet soap should be used without dilution. Type II paste toilet soap should be diluted 1:3 with water before use.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Net weight or volume per container.
(d) Marking required (see 5.3).

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

Preparing activity
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Orders for this publication are to be placed with General Services administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein.

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